

DRUMPANEL · REQUIREMENTS

DrumPanel requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/drumpanel		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3; status of every requirement from DMP-CAL-001 on the constructable design (DMP-DDR-002); R9 worded against the value-engineering target; R12 changed to three panels per drum
0.2	2026-10-03	Amish Chadha	TRL 2; targets firmed up; R10 guarding, R11 handling and R12 yield added; concept status for each requirement
0.1	2026-09-30	Amish Chadha	Initial scaffold

DrumPanel requirements

On paper, the constructable design meets ten of the twelve requirements, has one at risk and misses one. R7 (four drums an hour for two people) is not met: the estimate is about 2.0 drums an hour. R5 (flat within 10 mm over 1 m) is at risk because the right bending roll setting depends on each batch of drum steel and must be found by a trial pass. Every figure is from DMP-CAL-001 (docs/04-calcs/sizing.py); the tag in brackets is the line of its output.

Table 1. Requirements and their status at TRL 3.

ID	Requirement	Target	Verification (TRL 4)	Status at TRL 3
R1	No open flame in the process	Zero heat or flame steps from drum to flat sheet	Procedure review and observed trial	Met by design: purge, check, cut and roll are all cold; no grinder or torch touches a drum
R2	Vapour check before cutting	Check reliably flags a drum with flammable vapour at or above 10 % of the lower explosive limit (LEL)	Bump test of the bought detector with 50 % LEL gas; bench test with a calibrated gas	Met by design: a bought pumped LEL detector with its alarm set at 5 % LEL, half the target (DMP-DDR-001, D2)
R3	Remove ends cold	Both heads off a 208 L drum in under 10 minutes by one person	Timed trials on a batch of drums	Met on paper: about 3.9 min [H7]; crank force 12 to 27 N [C10 to C15]
R4	Open the side	Straight slit along the full body length, edge deviation at or below 5 mm	Measure the cut line on trial drums	Met by design: the shear head runs on a rail straight within 1 mm
R5	Flat sheet	Panels flat within 10 mm over 1 m after rolling	Straightedge and feeler check	At risk: needs the bending roll within 0.29 mm of the right height [E13]; a fixed setting can leave 188 mm of sag on other steel [E12]; set by a trial pass for each batch; the last 90 mm of each panel end is set by hand [E15, E16]
R6	Manageable effort	Crank force at or below 150 N at the handle	Spring scale on the handle during trials	Met on paper: end cutter 27 N, shear head 111 N at 1.5 mm wall, slip roll 20 N [C15, D15, E23]
R7	Throughput	At least 4 drums to flat sheet per hour for a two-person team	Timed half-day workshop trial	Not met: about 2.0 drums an hour [H5]
R8	Local build	All parts made or bought in a town with a welder and a lathe	Build by a partner fabricator from the drawings	Met by design: welding, drilling and turning only; the rolls need a lathe with 1.1 m between centres; discs, cutter wheel, bearings, gears and chain are bought
R9	Value engineering	Bench set parts cost at or below the USD 3,000 value-engineering target	Costed bill of materials	Met: estimated USD 2,794, USD 206 under the target [J1 to J3]
R10	Guarding	No finger reaches a roll nip, disc nip, gear or chain in normal use; guards need a tool to remove	ISO 13857 check of the guards as built	Met on paper: every slot 8 mm or less and 45 mm or more from its nip [G1 to G3]
R11	Handling	No lift over 25 kg by one person; a drum full of water is never lifted	Weigh parts; procedure review	Met with a two-person rule: heaviest piece the 34 kg cradle frame [I4]; a full drum (about 240 kg) is filled and drained in place [B2]
R12	Yield	Per drum: three flat panels about 233 x 1,800	Measure the output of trial drums	Met by design: panels 233, 234 and 233 x 1,800 mm; heads 552 mm [A1 to A3]

ID	Requirement	Target	Verification (TRL 4)	Status at TRL 3
		mm and two head discs about 550 mm across		

■ Assumptions

- Most drums reaching artisans held oils or similar contents that a water purge can clear enough to cut cold. Drums that did not are rejected at the door (DMP-PRB-001, Constraints).
- Artists accept painted panels, with paint removal as a separate cold step (DMP-DDR-001, D4).
- Drum body steel has a yield strength of 200 to 300 MPa and a wall of 0.8 to 1.2 mm; heads are 1.0 to 1.5 mm.
- The rolling hoops are cut out as two strips, so the slip roll never has to straighten them (DMP-DDR-002, P7).
- A pumped LEL detector, bump-tested each day, is good enough for a pass or fail check before cold cutting.

■ Safety

Safety: R1, R2, R10 and R11 are safety requirements. They are not traded against cost or throughput: where a cheaper or faster option would weaken them, the conservative option was taken (DMP-DEC-001). The bench set is an open engineering reference, never certified equipment.